

# Two lever positions, one load

*Autumn 1 · Week 4 · Lesson B · 40 minutes*

## Learning objective

Compare two lever pivot positions and explain the relationship between effort force and effort travel for the same load rise.

## Source lesson timing

| Elapsed minutes | Source stage     | Minutes |
|-----------------|------------------|---------|
| 0–32            | Independent Work | 32      |
| 32–36           | Lundy Loop       | 4       |
| 36–40           | Exit Ticket      | 4       |

Split slide times are added within their original source stages. Printed tasks and adaptations replace activity within these timings; they do not add extra lesson time.

## Equipment and preparation

- Adult-secured ruler/card lever and broad stable pivot
- Light secured load such as an eraser; two marked pivot positions
- Fixed effort point and small target load-height mark
- Optional suitable ruler/tape for travel; force meter only if appropriate for genuine measurements

TEACHING SEQUENCE

# Guide the lesson

*Use the source stage times; choose prompts appropriate to the pupil.*

## **Independent Work · 32 minutes**

Retrieve/check setup 4 min; compare middle and near-load pivots 20 min; check controls 2 min; repeat, explain and tidy 6 min. Same load, effort point and small target rise; qualitative effort unless properly measured.

## **Lundy Loop · 4 minutes**

Listen to pupil access/evidence feedback, agree one feasible change and explain limits.

## **Exit Ticket · 4 minutes**

Name pivot, explain lever benefit or force-distance trade-off; attempt then check.

## TEACHER ANSWERS

# Worked answers and expected evidence

*Use after an attempt. Accept equivalent correct explanations; record support given.*

## Gain and cost

For the same load and rise, near-load pivot increases effort-arm/load-arm ratio. Required effort force falls ideally while effort travel increases. Real friction and bending limit model; accept honest similar/uncertain observations.

### 1. Predict and prepare

Labels locate effort on bar, pivot as turning point and secured load. Controls: same load and position, same effort point and small target load rise. Prediction: nearer-load pivot may feel easier while effort end travels farther. A prediction is judged by reasoning, not whether expected observation occurs.

### 2. Record both positions

No fixed numbers expected. Ideal trend: near-load pivot requires less effort force and larger effort travel for equal load rise. Actual observations may be similar/uncertain due to friction, bending or method. Do not fabricate numbers or classify feeling as a force measurement in N. Check controls and record who operated/observed.

### 3. Check and explain

Corrected control should be specific. Explanation: effort arm becomes longer relative to load arm; smaller effort force ideally lifts same load, but effort end must move farther. Mechanism does not create energy. Friction and bending affect real model. No-equipment route states prediction/explanation, not practical observation.

### 4. Feedback and exit

Feedback has no prescribed response; adult agrees feasible change and explains limits. Pivot; longer effort arm relative to load arm lowers required effort force ideally; effort moves farther for same load rise and mechanism cannot create energy. Real losses may add effort.

## Misconceptions and corrections

- Feeling easier is a numerical force measurement: it is qualitative unless measured using an appropriate meter.
- Changing load rise is harmless: both positions must lift same load through same small rise.
- Diagram predicts exact real values: schematic has no measured scale/data.
- Mechanism creates energy: force gain has distance trade-off, plus real losses.

# Teach safely; keep the evidence

*The source record identifies this conversion; it is not a claim of publication.*

## Safety

- Use light secured load and controlled small lift; never launch objects.
- Adult checks pivot stability, attachment and work area before use.
- Hands clear of pivot; return all small parts and stop if kit shifts.
- Offer preassembled adult-operated model, observer/director route or labelled diagram explanation.

## Access and offline alternatives

- Direct/observe instead of touch, and dictate or point for records.
- No-kit route labels evidence as diagram explanation, never performed comparison.
- No internet/video/photography required; native diagram can be edited.

## Close and continue

Tuesday ends at 40 minutes. Preserve 32 workshop + 4 feedback + 4 exit. Next current Week 5 investigates fair-test variables with a trolley.

## Source and revision

Repository: [MattRoper1977/Lessons](https://github.com/MattRoper1977/Lessons)

Source path: [Science\\_Teesside/Grow/SCI\\_G\\_W4\\_Mechanisms.html](https://github.com/MattRoper1977/Lessons/blob/main/Science_Teesside/Grow/SCI_G_W4_Mechanisms.html)

Source file Git blob: [8aa3e1f001c3d3c944207d4eb44b93b1df4d7e46](https://github.com/MattRoper1977/Lessons/blob/main/Science_Teesside/Grow/SCI_G_W4_Mechanisms.html)

Current source link (mutable): [https://github.com/MattRoper1977/Lessons/blob/main/Science\\_Teesside/Grow/SCI\\_G\\_W4\\_Mechanisms.html](https://github.com/MattRoper1977/Lessons/blob/main/Science_Teesside/Grow/SCI_G_W4_Mechanisms.html)

Document revision: 2026-09-06

## Recorded conversion changes

- Current Tuesday structure kept: 32-minute workshop, 4-minute pupil feedback, 4-minute exit.
- Native editable diagram supports observation/directing and no-equipment explanation.
- Effort stays qualitative unless an appropriate force meter is actually used; no invented measurements.
- Same load, effort point and lift height are explicit controls.
- No publisher media copied; original logo resolution stays with release coordinator.